

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

1 Which expression is the correct factorisation of $8x^3 + 125$?

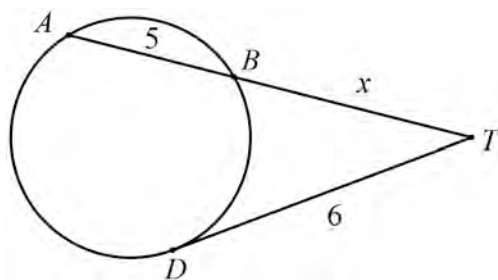
(A) $(2x+5)(4x^2+10x+25)$

(B) $(2x+5)(4x^2-10x+25)$

(C) $(2x-5)(4x^2+10x+25)$

(D) $(2x-5)(4x^2-10x+25)$

2



The line DT is a tangent to the circle at D and AT is a secant meeting the circle at A and B . Given that $DT = 6$, $AB = 5$ and $BT = x$, which of the following is the value of x ?

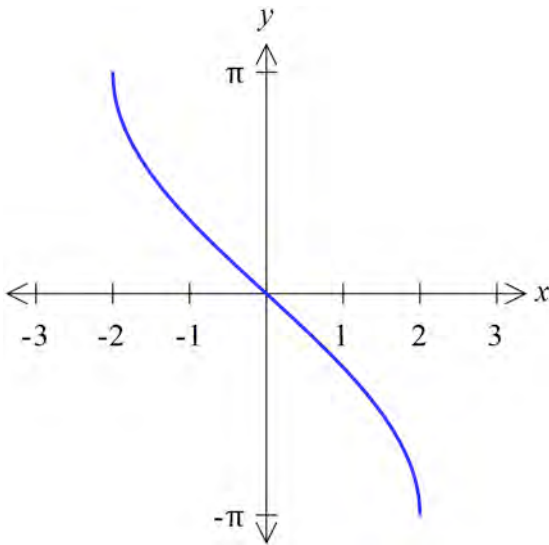
(A) $x = 4$

(B) $x = 5$

(C) $x = 6$

(D) $x = 9$

3 Which function does the following graph represent?



(A) $y = \frac{1}{2} \sin^{-1} 2x$

(B) $y = -\frac{1}{2} \sin^{-1} 2x$

(C) $y = 2 \sin^{-1} \frac{x}{2}$

(D) $y = -2 \sin^{-1} \frac{x}{2}$

4 What is the coefficient of x^5 in the expansion of $(2x+5)^8$?

(A) 1400000

(B) 224000

(C) 25000

(D) 4000

5 Which of the following is the solution to $\sin \theta = \cos 2\theta$, $0 \leq \theta \leq 2\pi$?

(A) $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

(B) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

(C) $\frac{4\pi}{3}, \frac{5\pi}{3}$

(D) $\frac{\pi}{3}, \frac{2\pi}{3}$

- 6 A is the point $(-5, 6)$ and B is the point $(4, 3)$. Which of the following points divides the interval AB externally in the ratio $3:2$?

- (A) $(-2, -21)$
(B) $(-27, 12)$
(C) $(22, -3)$
(D) $\left(\frac{2}{5}, 4\frac{1}{5}\right)$

- 7 What is the primitive of $\cos^2 x$?

- (A) $\frac{\sin 2x + 2x}{4} + C.$
(B) $\frac{1}{2} \sin 2x + x + C.$
(C) $\frac{4x - \sin 4x}{4} + C.$
(D) $x - \frac{1}{4} \sin 2x + C.$

- 8 What is the approximate size of the acute angle between the tangents to the curve $y = \ln(2x + 1)$ at the points where $x = 0$ and $x = \frac{1}{2}$?

- (A) 11°
(B) 18°
(C) 72°
(D) 79°

- 9** A particle is projected so that at any time t , its position (x,y) is given by

$$x = 36t, \quad y = 15t - \frac{1}{2}gt^2, \quad \text{where distances are in metres and time is in seconds.}$$

If θ is the angle of projection and V is the initial velocity, which of the following statements is correct?

- (A) $V = 36 \text{ m s}^{-1}$ and $\theta = \sin^{-1}\left(\frac{5}{13}\right)$
- (B) $V = 15 \text{ m s}^{-1}$ and $\theta = \cos^{-1}\left(\frac{12}{13}\right)$
- (C) $V = 39 \text{ m s}^{-1}$ and $\theta = \tan^{-1}\left(\frac{5}{13}\right)$
- (D) $V = 39 \text{ m s}^{-1}$ and $\theta = \tan^{-1}\left(\frac{5}{12}\right)$

- 10** Which expression is equivalent to $8\sin x - 15\cos x$?

- (A) $17\cos(x - 61^\circ 56')$
- (B) $17\cos(x + 61^\circ 56')$
- (C) $17\sin(x - 61^\circ 56')$
- (D) $17\sin(x + 61^\circ 56')$

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 marks) Use a **Separate** Booklet.

Marks

-
- (a) The polynomial equation $-2x^3 + 5x - 1 = 0$ has three roots α , β , and γ .
Evaluate the following

(i) $\beta\alpha + \beta\gamma + \alpha\gamma$.

1

(ii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$.

2

- (b) Solve $\frac{5-x}{x} < 3$

3

- (c) A particle moves such that its velocity is given by $v = x - 5$ m/s.
Show that the acceleration is the same as the velocity for all x .

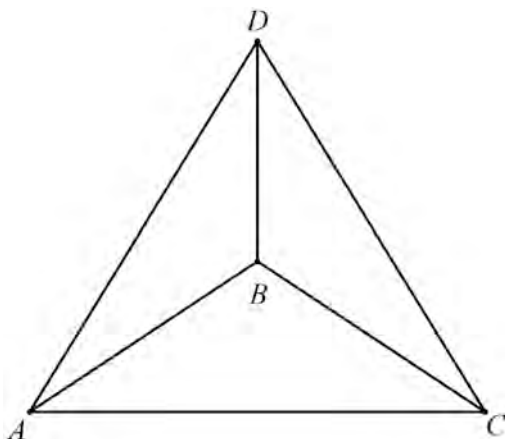
2

- (d) Write $\tan\left(\cos^{-1}\left(-\frac{1}{3}\right)\right)$ in the form $a\sqrt{b}$ where a and b are rational.

2

Question 11 continues on page 6

- (e) Jane and John are competing in a sailing boat race. Jane (C) can see the top of a cliff (D) that is 1500 m above sea level. The cliff is on a bearing of 359° from her position and the angle of elevation of the top of the cliff is 16° . John (A) can also see the cliff on a bearing of 059° with an angle of elevation of 23° . The base of the cliff (B) is at sea level.



- (i) Show that $\angle ABC = 60^\circ$. 1
- (ii) Copy the diagram and include all relevant information. 1
- (iii) Find the distance AC between the two sailing boats to the nearest metre. 3

End of Question 11

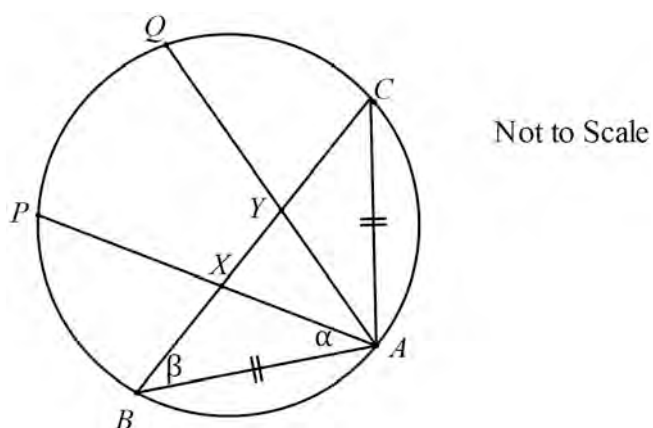
Question 12. (15 marks) Use a **Separate** Booklet.**Marks**

-
- (a) (i) Find the linear factors of $x^3 - 5x^2 + 8x - 4$. 2
- (ii) Hence solve $x^3 - 5x^2 + 8x - 4 > 0$. 1
- (b) Use the substitution $u = x^2 + 1$ to evaluate $\int_0^2 \frac{x}{(x^2 + 1)^3} dx$. 3
- (c) The volume of water in a tidal pool is given by the formula $V = 2 \cos \frac{3\pi}{x}$, 3
where x is the depth of water in the pool, in metres.
- Find the exact rate at which the depth of the pool will be increasing when the volume of water is increasing at $12 \text{ m}^3/\text{h}$ and the depth is 1.2 m .
- (d) A parabola is given by the parametric equations $x = t$ and $y = t^2$.
- (i) Sketch the parabola, and on your diagram mark the points P and Q which correspond to $t = -1$ and $t = 2$ respectively. 1
- (ii) Show that the tangents to the parabola at P and Q intersect at $R\left(\frac{1}{2}, -2\right)$. 2
- (iii) Let $T(t, t^2)$ be the point on the parabola between P and Q such that the tangent at T meets QR at the midpoint of QR . Show that the tangent at T is parallel to PQ . 3

End of Question 12

(a) If $\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{4-9x^2}} = k\pi$, find the value of k . 3

(b) In the circle below, $AB = AC$. Let $\angle PAB = \alpha$ and $\angle ABC = \beta$.



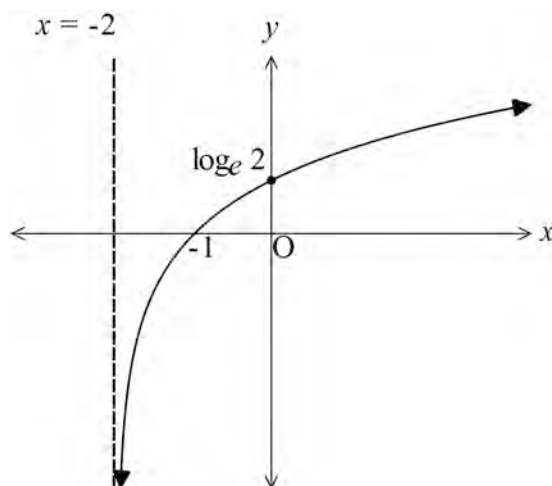
(i) Copy the diagram into your answer booklet and give a reason why $\angle PQB = \alpha$. 1

(ii) Prove $\angle AQB = \beta$. 1

(iii) Prove $XYQP$ is a cyclic quadrilateral. 2

Question 13 continues on page 9

- (c) The diagram below shows the graph of the function $y = f(x)$ where $f(x) = \ln(x+2)$.



- (i) Copy the diagram and on it draw the graph of the inverse function $y = f^{-1}(x)$ showing the intercepts on the axes and the equation of the asymptote. 2
- (ii) Show that the x coordinates of the points of intersection of the curves $y = f(x)$ and $y = f^{-1}(x)$ satisfy the equation $e^x - x - 2 = 0$. 2
- (iii) Show that the equation $e^x - x - 2 = 0$ has a root α such that $1 < \alpha < 2$. 2
- (iv) Use one application of Newton's method with an initial approximation $\alpha_0 = 1.2$ to find the next approximation for the value of α , giving your answer correct to one decimal place. 2

End of Question 13

- (a) Use mathematical induction to prove that for all integers $n \geq 1$,

3

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{(n+1)^2}\right) = \frac{n+2}{2n+2}.$$

- (b) In an experiment recording the number, N , of daisies in a given area of garden, it was found that the rate of change of N is given by

$$\frac{dN}{dt} = kN \left(1 - \frac{N}{2000}\right), \text{ where } k \text{ is a constant and } t \text{ is number of days.}$$

At the beginning of the experiment there were 1000 daises.

- (i) Verify that $N = \frac{2000e^{kt}}{1+e^{kt}}$ is a solution of the equation.

3

- (ii) If $N = 1500$ when $t = 10$ days, determine the value of t when $N = 1800$.

3

Question 14 continues on page 11

(c) The depth of water y metres in a tidal creek is given by

$$4\frac{d^2y}{dt^2} = 5 - y, \text{ the time } t \text{ being measured in hours.}$$

(i) Prove that the vertical motion of water is simple harmonic. 2
Find the centre of motion.

(ii) Write down the period T of the motion. 1

(iii) Given that $y = 1$ m at low tide and $y = 9$ m at high tide, and 1

that $y = a - b \cos nt$ is a solution of the equation $4\frac{d^2y}{dt^2} = 5 - y$,
write down the values of a , b and n .

(iv) If the low tide one day is at 1.00 pm, when is the earliest time that 2
a boat requiring 3 m of water can enter the creek?

Give your answer correct to the nearest minute.

End of Paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

2015 Extension 1 Trial Solutions

1. $8x^3 + 125$

$$= (2x)^3 + (5)^3$$

$$= (2x+5)((2x)^2 - 10x + 5^2)$$

$$(2x+5)(4x^2 - 10x + 25)$$

$\therefore$

B

2. $6^2 = (x+5) \cdot x$

$$x^2 + 5x - 36 = 0$$

$$(x+9)(x-4) = 0$$

$$x = -9, 4$$

But $x > 0 \therefore x = 4$

$\therefore$

A

3. When $x=2$ $-2\sin^{-1}\frac{2}{2} = -2 \times \pi/2 = -\pi \therefore$ D

4. General term $\binom{8}{r} (2x)^{8-r} 5^r$

For x^5 $r=3$

$$\therefore \text{coeff is } \binom{8}{3} 2^5 \cdot 5^3$$

$$= 224000$$

$\therefore$ B

5. $\sin \theta = \cos 2\theta$

$$= 1 - 2\sin^2 \theta$$

$$2\sin^2 \theta + \sin \theta - 1 = 0$$

$$(2\sin \theta - 1)(\sin \theta + 1) = 0$$

$$\therefore \sin \theta = 1/2 \text{ or } -1$$

$$0 \leq \theta \leq 2\pi$$

$$\theta = \pi/6, 5\pi/6, 3\pi/2$$

$\therefore$ B

6. min $\begin{matrix} x_1, y_1 \\ A(-5, 6) \end{matrix}$

$$(3, -2) \quad \begin{matrix} x_2, y_2 \\ B(4, 3) \end{matrix}$$

$$x = \frac{nx_1 + mx_2}{m+n} = \frac{(-2 \times -5) + (3 \times 4)}{3-2}$$

$$= 22$$

$\therefore$ C

7. $\int \cos^2 x \, dx$

$$\cos 2x = 2\cos^2 x - 1$$

$$\cos^2 x = 1/2 (\cos 2x + 1)$$

$$= \frac{1}{2} \left(\frac{1}{2} \sin 2x + x \right) + C$$

$$= \frac{\sin 2x + 2x}{4} + C$$

$\therefore$ A

8. Gradient is $\frac{dy}{dx} = \frac{2}{2x+1}$
Fr

When $x=0$ $m_1 = 2$

$x = 1/2$ $m_2 = 1$

$$\therefore \tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{2 - 1}{1 + 2 \times 1} \right| = \frac{1}{3}$$

$\alpha = 18^\circ$

$\therefore$ B

9. $x = (V \cos \alpha)t \therefore V \cos \alpha = 36$

$V \sin \alpha = 15$

$\therefore \tan \alpha = \frac{15}{36} = \frac{5}{12}$

$V^2 (\sin^2 \alpha + \cos^2 \alpha) = 15^2 + 36^2$

$\therefore V = \sqrt{15^2 + 36^2} = 39 \therefore$ D

10. $r \sin(x - \alpha) = r \sin x \cos \alpha - r \cos x \sin \alpha$

$r \cos \alpha = 8$

$\tan \alpha = 15/8$

$r \sin \alpha = 15$

$\alpha = 61^\circ 56'$

$r = \sqrt{8^2 + 15^2}$

$= 17$

$\therefore$ C

Question 11

$$a = -2$$

$$b = 0$$

$$c = 5$$

$$d = -1$$

$$\alpha + \beta + \gamma = -\frac{b}{a} = 0/-2 = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{5}{-2}$$

$$\alpha\beta\gamma = -\frac{d}{a} = -\frac{(-1)}{-2} = -1/2$$

a)

$$i) \quad \beta\alpha + \beta\gamma + \alpha\gamma = -5/2 //$$

1/1W

$$ii) \quad \alpha^{-1} + \beta^{-1} + \gamma^{-1}$$

$$= \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= -5/2 / -1/2$$

$$= 5 //$$

1 for correct
amalgamation

1 correct answer

$$b) \quad \frac{5-x}{x} < 3$$

③ - Completely
Correct

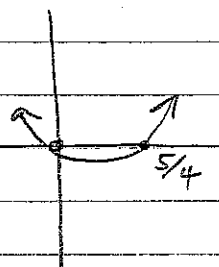
$$x(5-x) < 3x^2$$

$$5x - x^2 < 3x^2$$

$$4x^2 - 5x > 0$$

$$x(4x - 5) > 0$$

$$x < 0 \text{ or } x > 5/4 //$$



② - 1 error

① - 2 errors

$$c) \quad v = x - 5 \text{ m/s}$$

$$a_{\text{acc}} = \frac{dv}{dx} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} \frac{1}{2} (x-5)^2$$

$$= 2 \left(\frac{1}{2} \right) (x-5)^1 \cdot 1$$

$$= x - 5$$

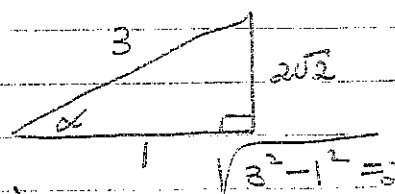
$$= v$$

② - correct method
and execution

① - 1 error

$\therefore$ acceleration is velocity for all x .

$$d) \quad \tan(\cos^{-1}(-1/3)) = -2\sqrt{2} //$$



$\cos^{-1}(-1/3)$ is obtuse $\therefore \tan(\cos^{-1}(-1/3)) < 0$

② - correct answer; accept $\sqrt{5}$

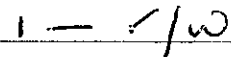
① - one error

Birds eye view



$$\therefore \angle ABC = 59^\circ + 1^\circ$$
$$= 60^\circ$$

ii)



① - 2 errors, or just finding AB and BC

Question 12.

a) i) Let $P(x) = x^3 - 5x^2 + 8x - 4$

$$P(1) = 1 - 5 + 8 - 4 = 0$$

$\therefore x-1$ is a factor

$$P(x) = (x-1)(x-2)^2 //$$

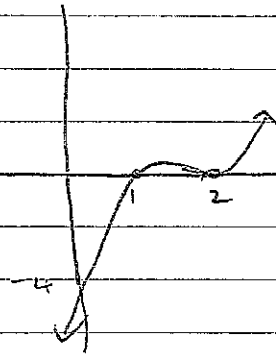
$$\begin{array}{r} x^2 - 4x + 4 \\ x-1 \overline{) x^3 - 5x^2 + 8x - 4} \\ \underline{-(x^3 - x^2)} \\ -4x^2 + 8x \\ \underline{-4x^2 + 4x} \\ 4x - 4 \\ \underline{4x - 4} \\ 0 \end{array}$$

ii) $x^3 - 5x^2 + 8x - 4 > 0$

$$(x-1)(x-2)^2 > 0$$

$$x > 1 \text{ except } x=2 //$$

or $1 < x < 2, x > 2 //$



b) $u = x^2 + 1$

$$\frac{du}{dx} = 2x \therefore du = 2x dx.$$

$$\frac{1}{2} du = x dx$$

When $x=0$ $u=1$

$x=2$ $u=5$

$$\int_0^2 \frac{x}{(x^2+1)^3} dx = \int_1^5 \frac{1}{u^3} \cdot \frac{1}{2} du$$

$$= \frac{1}{2} \int_1^5 u^{-3} du$$

$$= \frac{1}{2} \left[\frac{u^{-2}}{-2} \right]_1^5$$

$$= -\frac{1}{4} \left[\frac{1}{u^2} \right]_1^5$$

$$= -\frac{1}{4} \left(\frac{1}{25} - \frac{1}{1} \right)$$

$$= \frac{6}{25} //$$

$$12c) \quad V = 2\pi r^3 \frac{3\pi}{x}$$

Find $\frac{dx}{dt}$ when $\frac{dV}{dt} = 12$ and $x = 1.2$

$$\frac{dx}{dt} = \frac{dx}{dV} \cdot \frac{dV}{dt}$$

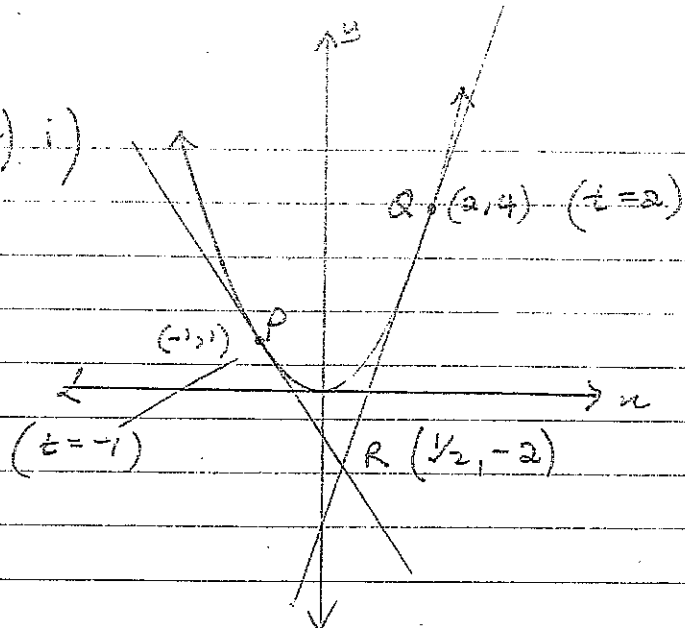
$$\begin{aligned} \frac{dV}{dx} &= \frac{d}{dx} 2\pi r^3 \left(\frac{3\pi}{x} \right) \\ &= \frac{-3\pi}{x^2} \cdot 2 \cdot \sin\left(\frac{3\pi}{x}\right) \\ &= \frac{6\pi}{x^2} \cdot \sin\left(\frac{3\pi}{x}\right) \quad (1) \end{aligned}$$

$$\frac{dx}{dV} = \frac{x^2}{6\pi \sin\left(\frac{3\pi}{x}\right)}$$

$$\begin{aligned} \text{When } x = 1.2 \quad \frac{dx}{dV} &= \frac{1.2^2}{6\pi \sin\frac{3\pi}{1.2}} = \frac{\frac{36}{25}}{6\pi \sin 5\pi/2} \quad (1) \\ &= \frac{36}{25} \times \frac{1}{6\pi} \end{aligned}$$

$$\begin{aligned} \therefore \frac{dx}{dt} &= \frac{6}{25\pi} \times 12 \\ &= \frac{72}{25\pi} \text{ m/h} \quad (1) \end{aligned}$$

a) i)



(1)

ii) $x = t, y = t^2$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= 2t$$

$\therefore$ At P, $\frac{dy}{dx} = -2$

Tangent at P is $y - 1 = -2(x - (-1))$

$$y = -2x - 1 \quad (1)$$

At Q, $\frac{dy}{dx} = 4$

Tangent at Q is $y - 4 = 4(x - 2)$

$$y = 4x - 4 \quad (2) \quad (1)$$

Solve (1) + (2) simultaneously to find R

$$-2x - 1 = 4x - 4$$

$$6x = 3$$

$$x = 1/2$$

When $x = 1/2$ $y = 4(1/2) - 4$ using (2)

$$y = 2 - 4 = -2 \quad (1)$$

$\therefore$ R is $(1/2, -2)$ //

iii) Gradient at $T(t, t^2)$ is $\frac{dy}{dx} = 2t$

Tangent at T is $y - t^2 = 2t(x - t)$

Midpoint of QR is $\left(\frac{2 + 1/2}{2}, \frac{4 + (-2)}{2} \right) = (5/4, 1)$

tangent at $(5/4, 1)$ lies on T $\therefore 1 - \left(\frac{t}{2}\right)^2 = 2t(5/4 - t) \quad (1)$

$$1 - t^2 = \frac{10t}{4} - 2t^2$$

$$4 - 4t^2 = 10t - 8t^2$$

$$4t^2 - 10t + 4 = 0$$

$$2t^2 - 5t + 2 = 0$$

$$2t^2 - 5t + 2 = 0$$

$$(2t - 1)(t - 2) = 0$$

$$t = \frac{1}{2} \text{ or } t = 2$$

but $t = 2$ is at Q $\therefore T$ is $(\frac{1}{2}, \frac{1}{4})$ ①

Gradient of tangent at T is $2t = 2 \times \frac{1}{2} = 1$

Gradient of PQ is $\frac{4-1}{2-1} = 1$

$\therefore$ Tangent at T is parallel to PQ // ①

Q13

$$a) \int_0^{2/3} \frac{dx}{\sqrt{4-9x^2}} = k\pi$$

$$\int_0^{2/3} \frac{dx}{\sqrt{9\left(\frac{4}{9} - x^2\right)}} = k\pi$$

$$\frac{1}{3} \int_0^{2/3} \frac{dx}{\sqrt{\left(\frac{2}{3}\right)^2 - x^2}} = k\pi$$

$$\frac{1}{3} \left[\sin^{-1} \frac{x}{(2/3)} \right]_0^{2/3} = k\pi$$

① coefficient $\frac{1}{3}$

$$\sin^{-1} \frac{2/3}{2/3} - \sin^{-1} \frac{0}{2/3} = 3k\pi$$

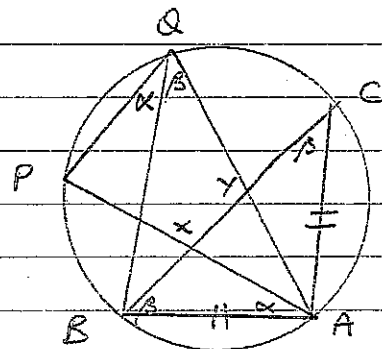
$$\pi/2 - 0 = 3k\pi$$

$$3k = 1/2$$

$$k = 1/6 //$$

①

b) i) $\angle PQB = \angle PAB = \alpha$
(Angles at the circumference subtended by arc PB are equal)



①

ii) $\angle ACB = \angle CBA = \beta$ (angles opposite equal sides in isosceles $\triangle ABC$)

$\angle AQB = \angle ACB$ (angles at the circumference subtended by arc AB are equal)

$$\therefore \angle AQB = \beta$$

①

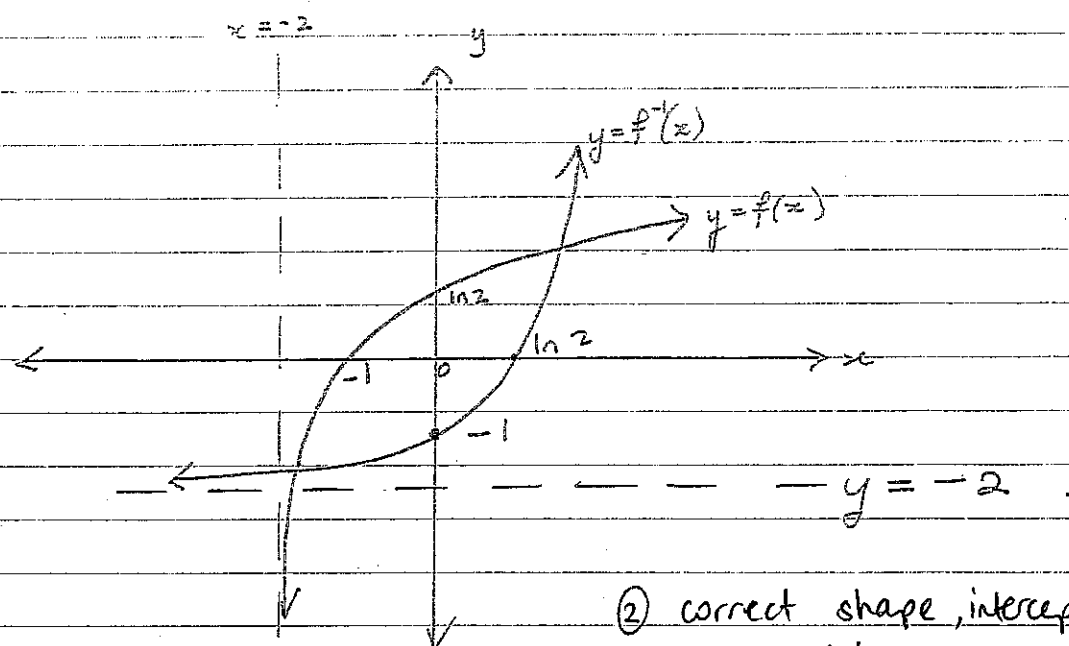
iii) $\angle YXA = \angle XBA + \angle XAB = \beta + x$ (exterior angle of $\triangle XAB$) ①

$\angle PQY = \angle PQB + \angle AQB = \alpha + \beta$ (adjacent angles, i+ii)

$$\therefore \angle PQY = \angle YXA = \alpha + \beta$$

$\Rightarrow$ $XYQP$ is a cyclic quadrilateral (exterior angle equals the opposite interior angle) ①

i)



② correct shape, intercepts, asymptote
- ① each error

ii) $y = f(x) : y = \ln(x+2)$ $f(x)$

$y = f^{-1}(x) : x = \ln(y+2)$
 $e^x = y+2$
 $y = e^x - 2$ $f^{-1}(x)$

$\therefore$ The intersection of the 2 curves is where :

$$\ln(x+2) = e^x - 2 \quad ①$$

Since $f(x)$ and $f^{-1}(x)$ intersect $\therefore x = e^x - 2$

$y=x \Rightarrow e^x - x - 2 = 0$ gives the ①
 x - co-ordinates of the points of intersection

iii) When $x=1$ $e^x - x - 2 = e^1 - 1 - 2 = e^1 - 3 < 0$ ①
 When $x=2$ $e^x - x - 2 = e^2 - 2 - 2 = e^2 - 4 > 0$

Since there is a change in sign of $e^x - x - 2$ from $x=1$ to $x=2$ and $e^x - x - 2$ is continuous between these values, there must be a root α , of $e^x - x - 2 = 0$ such that $1 < \alpha < 2$. ①

iv) Let $f(x) = e^x - x - 2$ $f(1.2) = e^{1.2} - 1.2 - 2$
 $f'(x) = e^x - 1$ $f'(1.2) = e^{1.2} - 1$

New approximation
 $a_1 = a_0 - \frac{f(a_0)}{f'(a_0)} = 1.2 - \frac{e^{1.2} - 3.2}{e^{1.2} - 1} \quad ①$
 $= 1.14822 \dots = 1.1 \text{ to 1 d.p.} \quad ①$

Q14

a) Let the statement be:

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{(n+1)^2}\right) = \frac{n+2}{2n+2}$$

Where $n=1$ LHS = $1 - \frac{1}{(1+1)^2} = 1 - \frac{1}{2^2} = \frac{3}{4}$ $\rightarrow$

$$\text{RHS} = \frac{1+2}{2(1)+2} = \frac{3}{4}$$

Since LHS = RHS, statement is true for $n=1$ (1)

Assume the statement is true for $n=k$ $k \geq 1$

ie $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2k+2}$

Show the statement is true for $n=k+1$

ie $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{(k+1)^2}\right) \left(1 - \frac{1}{(k+2)^2}\right) = \frac{k+3}{2k+4}$

$$\text{LHS} = \left(\frac{k+2}{2k+2}\right) \left(1 - \frac{1}{(k+2)^2}\right)$$

By assumption (1)

$$= \frac{k+2}{2(k+1)} \times \frac{(k+2)^2 - 1}{(k+2)^2}$$

must have
3 dot points
for mark.
not penalized if
forgotten

$$= \frac{(k+2)^2 - 1}{2(k+1)(k+2)}$$

$$= \frac{k^2 + 4k + 3}{2(k+1)}$$

$$= \frac{(k+3)(k+1)}{2(k+1)(k+2)}$$

$$= \frac{k+3}{2k+4}$$

$$= \text{RHS}$$

logical and shown
to the end

(1)

$\therefore$ The statement is true for $n=k+1$ if it is true for $n=k$ and since it is true for $n=1$, it is also true for $n=1+1=2$, $2+1=3$ and so on for all positive integer values of n .

Q14 continued

b) $\frac{dN}{dt} = kN \left(1 - \frac{N}{2000} \right)$ When $t=0$ $N=1000$

i) If $N = \frac{2000e^{kt}}{1+e^{kt}}$

then $\frac{dN}{dt} = \frac{(1+e^{kt}) \cdot 2000ke^{kt} - 2000e^{kt} \cdot ke^{kt}}{(1+e^{kt})^2}$

$= \frac{2000ke^{kt}((1+e^{kt}) - e^{kt})}{(1+e^{kt})(1+e^{kt})}$

① correctly differentiated

① must show how

$= k \left(\frac{2000e^{kt}}{1+e^{kt}} \right)$

$\frac{dN}{dt} = \frac{2000ke^{kt}}{(1+e^{kt})^2}$

Now $1 - \frac{N}{2000} = 1 - \frac{2000e^{kt}}{2000(1+e^{kt})}$

$= 1 - \frac{e^{kt}}{1+e^{kt}}$

$= \frac{1+e^{kt} - e^{kt}}{1+e^{kt}}$

$= \frac{1}{1+e^{kt}}$

① must show how.

$\therefore \frac{dN}{dt} = kN \left(\frac{1}{1+e^{kt}} \right)$

$= kN \left(1 - \frac{N}{2000} \right)$

$\Rightarrow N = \frac{2000e^{kt}}{1+e^{kt}}$ is a solution of the equation

iii) When $t=10$ $N=1500$ $\therefore 1500 = \frac{2000e^{10k}}{1+e^{10k}}$

$1500 + 1500e^{10k} = 2000e^{10k}$

$500e^{10k} = 1500$

$e^{10k} = 3$ ①

$10k = \ln 3$

$k = \frac{1}{10} \ln 3$ ①

$\therefore$ When $N=1800$

$1800 = \frac{2000e^{kt}}{1+e^{kt}}$

$$1800 + 1800e^{kt} = 2000e^{kt}$$

$$(k = \frac{\ln 3}{10})$$

$$2000e^{kt} = 1800$$

$$e^{kt} = \frac{9}{10}$$

$$kt = \ln \frac{9}{10}$$

$$t = \frac{1}{k} \ln \frac{9}{10}$$

$$= \frac{10 \ln \frac{9}{10}}{\ln 3}$$

$$= \boxed{20} \text{ days} \quad \text{--- (1)}$$

as value of t asked for.

c) i) $4 \frac{d^2 y}{dx^2} = 5 - y$

For SHM $\ddot{x} = -n^2(x - b)$ where b is the centre of motion.

$$4 \ddot{y} = 5 - y$$

$$4 \ddot{y} = -(y - 5) \quad \text{(1)}$$

$$\ddot{y} = -\frac{1}{4}(y - 5)$$

which is SHM with centre 5. (1)

ii) $T = \frac{2\pi}{n} = \frac{2\pi}{(1/2)} = \boxed{4\pi} \quad \text{(1)}$

iii) $y = a - b \cos nt$
 $= -b \cos nt + a$

$y = 1 \leftarrow \text{cot M.} = 5$
 $y = 9$

$y = -b \cos nt$ begins at $-b + a$. $\therefore$ amplitude is 4
 $\therefore -4 + a = 1 \Rightarrow a = 5$. $\therefore b = 4$

$$\therefore \boxed{a = 5 \quad b = 4 \quad n = 1/2} \quad \text{(1)}$$

(N) For $y = 3$

$$3 = 5 - 4 \cos \frac{t}{2}$$

$$\frac{1}{2} = \cos \frac{t}{2}$$

$$\frac{t}{2} = \frac{\pi}{3}, \frac{2\pi}{3} \text{ etc}$$

$$t = \frac{2\pi}{3}, \frac{4\pi}{3} \text{ etc}$$

* needed to round up!!

$\therefore$ earliest time for boat to enter the creek

is when $t = \frac{2\pi}{3} \text{ h} = 2 \text{ h } 5' 39.82''$

i.e. $3:06 \text{ pm} \quad \text{(1)} = 2 \text{ h } 6' \text{ to nearest min.}$